

# Shan Yang

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## SUMMARY

Senior Applied Scientist | Video Generation, Multimodal Modeling, GenAI Systems

## EXPERIENCE

### Senior Applied Scientist Tech Lead

Amazon Prime Video

June 2025 – present, Sunnyvale, CA

- Established Prime Video's post-training pipeline (LoRA, ControlNet, Test-Time-Training, Self-forcing, Diffusion-forcing) enabling controllable, consistent, and context-preserving generative video, forming the adaptation layer for future GenAI creative systems;
- Driving foundational research on conditional video generation, advancing auto-regressive diffusion, test-time conditioning mechanisms, and controllable keyframe video synthesis using diffusion forcing;
- Author of multiple internal publications shaping the long-term GenAI video roadmap for Prime Video.

### Senior Applied Scientist Tech Lead

Amazon Search

September 2021 – June 2025

- Led and launched Amazon Video Search on the Amazon product search page, building a new multi-modal customer experience adopted by millions of users;
- Developed the Video Search Ranker, combining video semantic representations with text, image, and product metadata features through a unified multi-modal architecture optimized for large-scale retrieval and production search workloads;
- Architected the end-to-end Video Search system: retrieval, ranking, multimodal fusion, and inference;
- Led multimodal LLM efforts for captioning and VQA;
- Owned full-stack development, deployment, and A/B testing;
- Granted an Amazon patent for multimodal ranking innovations.

### Senior Research Software Engineer

Google Research

June 2017 – September 2021, Mountain View

- Led multimodal video content creation research, including music-conditioned 3D dance generation and video-audio-motion modeling;
- Conducted core 3D vision research, focusing on monocular video pose estimation and structured motion understanding;
- Served as the primary owner and major contributor for Google's 2D/3D reconstruction ML models, covering architecture design, training, and evaluation;
- Built an end-to-end monocular video analysis system used for biological and biomechanical research;
- Major contributor to a Google patent in multimodal generative modeling and vision-based reconstruction.

### Research Intern

Google Research

June 2016 – August 2016, Mountain View

- Engineered a variational autoencoder framework for 3D human reconstruction, enabling the generation of accurate body models from limited input data.

### Software Engineer Intern

Samsung Research America, San Jose

June 2014 – August 2014, San Jose

- Integrated automated testing procedures for graphics hardware modules, enhancing reliability and throughput of validation workflows.

## EDUCATION

### Ph.D. Computer Science

University of North Carolina at Chapel Hill • NC • 2012 – 2017

### B.E. Software Engineering

Shanghai Jiao Tong University • China • 2008 – 2012

## PUBLICATIONS

### Linear Recurrent Model for Test-Time Reasoning via Linear Quadratic Control

Peihao Wang, Changlong Yu, Rene Vidal, **Shan Yang** • Pending Publication • 2025

- We derive a novel neural layer that performs test-time planning to embed forward-looking computation directly within the model. We design a novel CUDA implementation that parallelizes this optimal planning on GPUs efficiently. By incorporating it as an adapter into pretrained LLMs, our

approach enhances mathematical reasoning performance, compared with LLaMA3 8B, +35.8% on Math-500, +6.7% on AIME24, +10% on AIME25, +9.3% on AMC.

### **ClipDreamer: Towards Faithful and Visually-Grounded Product Video Generation**

Zun Wang, Kenan Deng, Mohit Bansal, **Shan Yang** · Pending Publication · 2025

- We propose to use the Test-Time-Training layer to inject video clip conditions to drive faithful motion and appearance product-centric video generation.

### **Teacher Diffusion Forcing: New Training Paradigm for Video Generation**

Peihao Wang, **Shan Yang** · Pending Publication · 2025

- We introduce Teacher-Diffusion Forcing, a unified training paradigm that integrates the strengths of both teacher forcing and diffusion forcing to improve the efficiency and quality of video generation.

### **SoT: Schema-of--Thought for Robust Video Question Answering**

Xijun Wang, **Shan Yang** · Pending Publication · 2024

- We propose to apply the Schema Theory to dynamically prompt the multi-modal models to maximally reduce the poor perception performance problem. ChatGPT-4o with SoT achieved +13.3% improvement, Claude3.5 Sonnet with SoT achieved +28% improvement on MMVP. And we found SoT consistently improved the performance on public hallucination benchmarks.

### **VILA: Efficient video-language alignment for video question answering**

Xijun Wang, Junbang Liang, Chun-Kai Wang, Kenan Deng, Yu Lou, Ming C Lin, **Shan Yang** · European Conference on Computer Vision (ECCV) · 2024

### **ICAR: Image-based Complementary Auto Reasoning**

Xijun Wang, Anqi Liang, Junbang Liang, Ming Lin, Yu Lou, **Shan Yang** · AAAI · 2023

### **MeSa: Masked, Geometric, and Supervised Pre-training for Monocular Depth Estimation**

Muhammad Osama Khan, Junbang Liang, Chun-Kai Wang, **Shan Yang**, Yu Lou · NeurIPS 2023 WorkshopSSLTheoryPractice · 2023

### **Attention Bottlenecks for Multimodal Fusion**

Arsha Nagrani, **Shan Yang**, Anurag Arnab, Aren Jansen, Cordelia Schmid, Chen Sun · Neural Information Processing Systems (NeurIPS) · 2021

### **Optical Mouse: 3D Mouse Pose From Single-view Video**

**Shan Yang\***, Bo Hu\*, David A. Ross, Avneesh Sud, Yi Liu, Graham Ruby, Bryan Seybold · CVPR 2021 CV4Animal Workshop · 2021

### **Learn to Dance with AIST++: Music Conditioned 3D Dance Generation**

**Shan Yang\***, Ruilong Li\*, David A. Ross, Angjoo Kanazawa · International Conference on Computer Vision (ICCV) · 2021

### **Physics-Inspired Garment Recovery from a Single-View Image**

**Shan Yang**, Zherong Pan, Tanya Amert, Ke Wang, Licheng Yu, Tamara Berg, Ming C. Lin · ACM Transaction on Graphics (TOG) · 2018

### **Learning-based Cloth Material Recovery from Video**

**Shan Yang**, Junbang Liang, Ming C. Lin · International Conference on Computer Vision (ICCV) · 2017

### **Modeling Context in Referring Expressions**

Lichens Yu, Patric Poirson, **Shan Yang**, Alex Berg, Tamara Berg · European Conference on Computer Vision(ECCV) · 2016

### **Classification of Prostate Cancer Grades and T-Stages based on Tissue Elasticity Using Medical Image Analysis**

**Shan Yang**, Vladimir Jojic, Jun Lian, Ronald Chen, Hongtu Zhu and Ming C. Lin · International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) · 2016

### **Bayesian Estimation of Non-Rigid Mechanical Parameters Using Temporal Sequences of Deformation Samples**

**Shan Yang**, Ming C. Lin · IEEE International Conference on Robotics and Automation (ICRA) · 2016

### **Material Cloning: Acquiring Elasticity Parameters from Images**

**Shan Yang**, Ming C. Lin · IEEE Transactions on Visualization and Graphics (TVCG) · 2016

### **Simultaneous Estimation of Elasticity for Multiple Deformable Bodies**

**Shan Yang**, Ming C. Lin · Journal Computer Animation and Virtual Worlds (CAVW) · 2015

## **Patents**

Monitoring Animal Pose Dynamics from Monocular Images, **Shan Yang**, Bo Hu, Kevin Murphy, David Ross, 2022

Image-based Complementary Item Recommendation Algorithm, **Shan Yang**, Xijun Wang, Ming Lin, Anqi Liang, 2022

Auto Design: ML-based Furniture Layout Recommendation Algorithm, **Shan Yang**, Kenan Deng, Yu Lou, 2022

VLAP: Efficient Video-Language Alignment via Frame Prompting and Temporal Distilling, **Shan Yang**, Xijun Wang, 2023